

Math 526, S05
Exam 1

Name:
SSN: xxx-xx-

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	10	
2	20	
3	20	
4	20	
5	15	
6	15	
Total	100	

Good Luck !

Problem 1. (10 points) Let $\mathbf{A} = \begin{bmatrix} 0 & -2 & 5 \\ -2 & 1 & -3 \\ 4 & 2 & -1 \\ -1 & 2 & 3 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 1 & -2 & 3 \\ -1 & 4 & -3 \end{bmatrix}$.

a) Which one of the products $\mathbf{AB}^t$ and $\mathbf{A}^t\mathbf{B}$ is possible? Explain.

Solution: Since $\mathbf{A}$ is a 4×3 matrix and $\mathbf{B}^t$ is a 3×2 matrix, $\mathbf{AB}^t$ is possible. Since $\mathbf{A}^t$ is a 3×4 matrix and $\mathbf{B}$ is a 2×3 matrix, $\mathbf{A}^t\mathbf{B}$ is not possible.

b) Compute the product from a) that is possible.

Solution:

$$\mathbf{AB}^t = \begin{bmatrix} 19 & -23 \\ -13 & 15 \\ -3 & 7 \\ 4 & 0 \end{bmatrix}$$

Problem 2. (20 points)

a) Find two 2×2 matrices $\mathbf{A}$ and $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{0}_{2 \times 2}$, but $\mathbf{A} \neq \mathbf{0}_{2 \times 2}$ and $\mathbf{B} \neq \mathbf{0}_{2 \times 2}$.

Solution: There are infinite solutions. For example,

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

b) Let $\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix}$. Is the matrix $\mathbf{I}_{4 \times 4} - \mathbf{u}\mathbf{v}^t$ nonsingular? Explain.

Solution: We compute

$$\mathbf{v}^t \mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 & -1 \end{bmatrix} = 3$$

According to Sherman-Morrison Theorem ($\mathbf{v}^t \mathbf{u} \neq 1$), we know that $\mathbf{A}$ is nonsingular.

Problem 3. (20 points)

a) Consider the 2×3 matrices

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 2a_{11} - a_{21} & 2a_{12} - a_{22} & 2a_{13} - a_{23} \\ 5a_{11} - 2a_{21} & 5a_{12} - 2a_{22} & 5a_{13} - 2a_{23} \\ -3a_{11} + a_{21} & -3a_{12} + a_{22} & -3a_{13} + a_{23} \end{bmatrix}.$$

Find a 3×2 matrix $\mathbf{B}$ so that $\mathbf{A}$ multiplied by $\mathbf{B}$ equals $\mathbf{C}$. From which side should you multiply $\mathbf{A}$ by $\mathbf{B}$?

Solution: Let

$$\mathbf{B} = \begin{bmatrix} 2 & -1 \\ 5 & -2 \\ -3 & 1 \end{bmatrix}$$

Then

$$\mathbf{BA} = \mathbf{C}$$

b) How many flops does it take to compute the product of a tridiagonal $n \times n$ matrix and a column n -vector if multiplication by zero is not executed? Explain.

Solution:

$$3 + 5(n - 2) + 3 = 5n - 4$$

Problem 4. (20 points) Let $\mathbf{A} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 1 & -1 \\ -2 & 4 & -8 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ 10 \end{bmatrix}$.

a) Find the $\mathbf{L} - \mathbf{U} - \mathbf{P}$ factorization of $\mathbf{A}$ using the strategy of partial pivoting.

$$\begin{array}{ccc} \begin{bmatrix} \\ \\ \end{bmatrix} & \begin{bmatrix} -1 & -1 & -1 \\ -1 & 1 & -1 \\ -2 & 4 & -8 \end{bmatrix} & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \begin{bmatrix} \\ \\ \end{bmatrix} & \begin{bmatrix} -2 & 4 & -8 \\ -1 & 1 & -1 \\ -1 & -1 & -1 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \\ \begin{bmatrix} 1/2 \\ 1/2 \\ \end{bmatrix} & \begin{bmatrix} -2 & 4 & -8 \\ 0 & -1 & -5 \\ 0 & -3 & -5 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \\ \begin{bmatrix} 1/2 \\ 1/2 \\ \end{bmatrix} & \begin{bmatrix} -2 & 4 & -8 \\ 0 & -3 & 3 \\ 0 & -1 & 3 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \\ \begin{bmatrix} 1/2 \\ 1/2 & 1/3 \end{bmatrix} & \begin{bmatrix} -2 & 4 & -8 \\ 0 & -3 & 3 \\ 0 & 0 & 2 \end{bmatrix} & \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \end{array}$$

So we have

$$\mathbf{L} = \begin{bmatrix} 1 & & & \\ 1/2 & 1 & & \\ 1/2 & 1/3 & 1 & \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} -2 & 4 & -8 \\ 0 & -3 & 3 \\ 0 & 0 & 2 \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

b) Using the $\mathbf{L} - \mathbf{U} - \mathbf{P}$ factorization from a) to solve $\mathbf{Ax} = \mathbf{b}$.

Solution: First we solve $\mathbf{Ly} = \mathbf{Pb}$:

$$\begin{bmatrix} 1 & & \\ 1/2 & 1 & \\ 1/2 & 1/3 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 2 \\ 0 \end{bmatrix}$$

We get

$$y_1 = 10, \quad y_2 = -3, \quad y_3 = -4.$$

Then we solve $\mathbf{Ux} = \mathbf{y}$:

$$\begin{bmatrix} -2 & 4 & -8 \\ 0 & -3 & 3 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ -3 \\ -4 \end{bmatrix}$$

We get

$$x_1 = 1, \quad x_2 = -1, \quad x_3 = -2.$$

Problem 5. (15 points) Let $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}$.

a) Explain why $\mathbf{A}$ is invertible.

Solution: Since $\mathbf{A}$ is a lower triangular matrix and there are no zero's in the diagonal of $\mathbf{A}$, $\mathbf{A}$ is invertible.

b) Find the inverse matrix of $\mathbf{A}$.

Solution: The first column of $\mathbf{A}^{-1}$ is computed by solving

$$\mathbf{A}\mathbf{x} = \mathbf{e}_1$$

i.e.,

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

We get

$$x_1 = 1, \quad x_2 = -2, \quad x_3 = 7.$$

The second column of $\mathbf{A}^{-1}$ is computed by solving

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

We get

$$x_1 = 0, \quad x_2 = 1, \quad x_3 = -3.$$

The third column of $\mathbf{A}^{-1}$ is computed by solving

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

We get

$$x_1 = 0, \quad x_2 = 0, \quad x_3 = 1.$$

So

$$\mathbf{A}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 7 & -3 & 1 \end{bmatrix}$$

Problem 6. (15 points)

$$\text{Let } \mathbf{A} = \begin{bmatrix} 1 & -1 & & & \\ -1 & 3 & 4 & & \\ & 2 & 5 & -1 & \\ & & 2 & -4 & 4 \\ & & & -2 & 8 \end{bmatrix}.$$

a) Is $\mathbf{A}$ column diagonally dominant? Explain.

Solution: Since in the third column, $5 < 4 + 2 = 6$.

b) Find the $\mathbf{L} - \mathbf{U}$ factorization of the tridiagonal matrix $\mathbf{A}$ using the special procedure for tridiagonal matrices.

Solution:

$$\mathbf{L} = \begin{bmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{bmatrix}$$

$$\mathbf{U} = \begin{bmatrix} 1 & -1 & & & \\ & 2 & 4 & & \\ & & 1 & -1 & \\ & & & -2 & 4 \\ & & & & 4 \end{bmatrix}.$$